

Reference-Assisted Phase Retrieval for Holographic MIMO Reception: Unique Channel and Finite-Alphabet Symbol Recovery

Abstract—This paper investigates reference-assisted signal recovery for holographic reconfigurable intelligent surface (RIS) receivers with intensity-only measurements. A known reference wave interferes with the received field before square-law detection, thereby converting the resulting inverse problem into affine phase retrieval and removing the global-phase ambiguity. We derive an exact secant identity, continuous-channel identifiability conditions, and an energy-domain minimum-distance criterion for finite-alphabet detection. A four-phase reference construction further yields an explicit stability lower bound. Based on these results, a two-stage receiver is developed: a damped Gauss–Newton method jointly estimates the directions of arrival and complex path gains, while a reference-assisted Gerchberg–Saxton detector recovers data symbols, with maximum-likelihood detection serving as a benchmark. A Cramér–Rao lower bound characterizes the attainable angular accuracy. Simulations verify the predicted effects of reference design, training length, array size, and measurement diversity on channel estimation and symbol detection.

Index Terms—Reconfigurable intelligent surface, holographic receiver, phase retrieval, signal recovery, intensity-only measurements.

I. INTRODUCTION

Reconfigurable intelligent surfaces (RISs) provide a programmable electromagnetic interface for controlling wireless propagation with low-complexity surface elements. Their operating principles, communication models, and potential gains in coverage and energy efficiency have been extensively studied [1]–[9]. Beyond conventional reflecting architectures, densely sampled large intelligent and holographic surfaces can act as spatially continuous apertures, offering high spatial resolution and new wave-domain processing capabilities [10]–[13]. Recent studies have further explored holography-inspired surface control, wave-domain DOA estimation, and metasurface-enabled sensing and localization [14]–[18]; wavenumber-domain processing provides a complementary framework for modeling and processing quasi-continuous holographic apertures [19].

A central implementation challenge is that acquiring the complex field at every surface element conventionally requires coherent radio-frequency chains, local-oscillator distribution, and in-phase/quadrature sampling. Power-detector-based architectures instead retain only intensity measurements and can substantially simplify the receiving hardware [20], [21]. Square-law or direct detection is well established in optical and infrared communication systems [22], [23]; when used for complex-valued radio signals, however, it discards the phase that carries essential channel and symbol information. The resulting inverse problem is therefore a form of phase retrieval.

Phase retrieval has a long history in optics and signal processing [24], [25]. Convex lifting and semidefinite relaxations provide recovery guarantees at the cost of high-dimensional optimization [26]–[28], whereas alternating-minimization and gradient-based methods offer more scalable nonconvex alternatives [29], [30]. Recent deep-unfolding approaches further combine interpretable iterative structure with learned signal priors [31]. In parallel, fundamental results have characterized the number and structure of measurements required for uniqueness and stable recovery [32]–[36]. These results clarify the generic phaseless inverse problem, but its unavoidable global-phase ambiguity remains incompatible with the direct recovery of absolute communication symbols.

A known reference wave can remove this ambiguity by interfering with the unknown field before square-law detection. This principle originates from holography, where the object and reference waves form an intensity pattern that encodes their relative phase [37], [38]; controlled reference phases can provide additional complex-field information [39]. Algebraically, the reference converts homogeneous magnitude measurements into affine magnitude measurements, connecting the receiver considered here to affine phase retrieval [40]. Unlike a conventional coherent local oscillator used after antenna combining, the reference in this architecture participates directly in the element-wise interference pattern.

Phaseless recovery has also appeared in communication problems such as OFDM channel estimation [41]. Meanwhile, RIS channel acquisition and codebook design have been studied for conventional complex-baseband observations [42]–[44]. These works establish useful estimation and training mechanisms, but they do not resolve the identifiability of finite-alphabet symbols from reference-assisted intensity measurements. In particular, generic continuous-signal guarantees do not directly describe when two constellation vectors produce identical or nearly identical energy patterns under the coupled effects of the reference wave, waveform sampling, and RIS array manifold.

This paper addresses that gap through a unified reference-assisted phase-retrieval framework for holographic RIS receivers. We first separate field-domain injectivity from reference-induced intensity separation and derive conditions under which finite-alphabet symbols are uniquely represented by the observed energies. We then introduce an energy-domain minimum-distance metric that quantifies noise robustness and exposes the joint roles of reference diversity and RIS sampling. Finally, we develop practical channel-estimation and symbol-detection algorithms whose performance can be interpreted through these identifiability and separation properties.

The main contributions of this paper are summarized as follows:

- We establish a reference-assisted holographic RIS receiver model, in which complex-valued communication symbols are recovered from intensity-only measurements through the interference between the unknown signal and a known reference wave.
- We derive discrete identifiability conditions for reference-assisted phase retrieval over finite constellations and reveal the joint impact of the reference wave, the pulse-shaping matrix, and the RIS array manifold on symbol uniqueness.
- We introduce an energy-domain minimum distance metric to characterize the stability of symbol recovery and to guide reference-wave and RIS measurement design.
- We develop and compare maximum-likelihood and low-complexity phase retrieval based detectors, demonstrating that proper reference-wave diversity significantly improves symbol distinguishability and detection performance.

II. SYSTEM MODEL

We consider a holographic RIS receiver in which L single-antenna users transmit complex baseband symbols to a surface composed of multiple receiving elements. Each element senses the intensity of the interference pattern formed by the received field and a known reference wave generated by a local oscillator [21]. The receiver first estimates the complex channel parameters and directions of arrival (DOAs) and then uses the resulting channel estimate to recover the transmitted symbols.

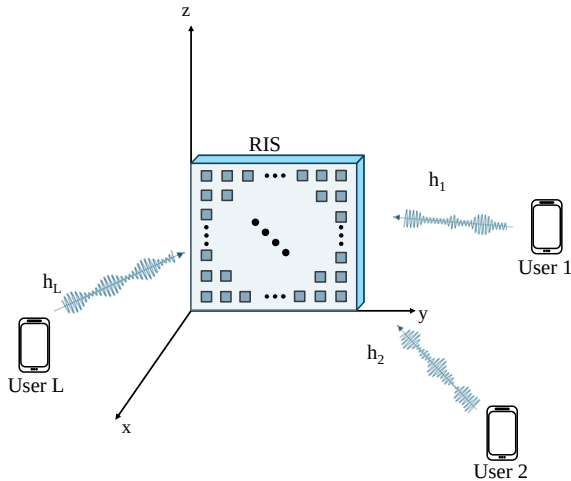


Fig. 1. System model of the holographic RIS receiver with intensity-only measurements.

A. Transmitter Model

The K symbols transmitted by user l in one signaling block are collected in $\mathbf{s}_l = [s_{l,0}, \dots, s_{l,K-1}]^T \in \mathcal{S}^K$, where $\mathcal{S} = \{c_0, \dots, c_{M_{\text{mod}}-1}\} \subset \mathbb{C}$ is an equiprobable constellation

normalized to $\frac{1}{M_{\text{mod}}} \sum_{q=0}^{M_{\text{mod}}-1} |c_q|^2 = \mathbb{E}\{|s_{l,k}|^2\} = 1$. After pulse shaping, the complex baseband waveform is

$$x_l(t) = \sqrt{P_l} \left(\sum_{k=0}^{K-1} s_{l,k} p(t - kT_s) + \xi_l q(t) \right), \quad (1)$$

Here, P_l is the transmit power, $p(t)$ is the pulse-shaping filter, T_s is the symbol duration, and the optional deterministic term $\xi_l q(t)$ is set to zero when no transmit-side pilot or bias is required. The corresponding real passband waveform radiated by user l is

$$x_l^{\text{RF}}(t) = \sqrt{2} \text{Re}\{x_l(t) e^{j2\pi f_c t}\}, \quad (2)$$

where f_c is the carrier frequency. Under the usual carrier-period averaging, the factor $\sqrt{2}$ makes the average power of $x_l^{\text{RF}}(t)$ equal to the envelope power $|x_l(t)|^2$. Thus, $x_l(t)$ contains the pulse-shaped amplitude and phase information, while the carrier term translates that waveform to RF for propagation.

B. Far-Field Channel Model

As shown in Fig. 1, the holographic RIS is an $N_y \times N_z$ uniform planar array on the yz -plane, with $N_r = N_y N_z$ elements. The (n, m) -th element is located at $\mathbf{r}_{n,m} = [0, y_n, z_m]^T$, where $y_n = (n - (N_y + 1)/2)d_y$ and $z_m = (m - (N_z + 1)/2)d_z$. The centered coordinates introduce no loss of generality because a change of origin produces only a user-dependent phase that can be absorbed into the complex gain.

Under far-field propagation, user l produces a plane wave with polar angle θ_l measured from the array broadside and azimuth angle ϕ_l measured from the positive y -axis in the yz -plane. With $k_0 = 2\pi/\lambda$, the relevant tangential wavenumbers are $k_{l,y} = k_0 \sin \theta_l \cos \phi_l$ and $k_{l,z} = k_0 \sin \theta_l \sin \phi_l$. The phase at element (n, m) is

$$\psi_{n,m}^{(l)} = \mathbf{k}_l^T \mathbf{r}_{n,m} = k_{l,y} y_n + k_{l,z} z_m. \quad (3)$$

The far-field amplitude is taken as constant across the aperture, while path loss, shadowing, and the user-dependent phase are absorbed into $\alpha_l \in \mathbb{C}$. Accordingly,

$$\mathbf{h}_l = \alpha_l \mathbf{v}_l, \quad [\mathbf{v}_l]_{\iota(n,m)} = e^{j\psi_{n,m}^{(l)}}, \quad (4)$$

where $\iota(n, m) = (n - 1)N_z + m$. Collecting all users gives

$$\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_L] = \underbrace{[\mathbf{v}_1, \dots, \mathbf{v}_L]}_{\mathbf{V}} \underbrace{\text{diag}(\alpha_1, \dots, \alpha_L)}_{\mathbf{\Gamma}} = \mathbf{V}\mathbf{\Gamma}, \quad (5)$$

which separates the directional array response from the complex path gains.

C. Reference-Assisted Holographic Reception Model

Let $\mathbf{x}(t) = [x_1(t), \dots, x_L(t)]^T$. At RIS element i , the locally generated reference is

$$b_i^{\text{RF}}(t) = \sqrt{2} \text{Re}\{b_i e^{j2\pi f_c t}\}, \quad b_i = \rho_i e^{j\chi_i}, \quad (6)$$

where ρ_i and χ_i are its prescribed amplitude and phase. Let $\mathbf{b} = [b_1, \dots, b_{N_r}]^T$ collect the reference coefficients, which are carrier aligned and phase stable over each detection interval.

Under the narrowband model, the received user field has envelope $[\mathbf{H}\mathbf{x}(t)]_i$. Its superposition with (6) gives

$$r_i^{\text{RF}}(t) = \sqrt{2} \text{Re}\{([\mathbf{H}\mathbf{x}(t)]_i + b_i) e^{j2\pi f_c t}\}. \quad (7)$$

Here, $\mathbf{H}\mathbf{x}(t)$ contains the propagation response and unknown user phases, whereas b_i is receiver controlled. The reference-assisted interference is therefore formed in the analog RF domain before square-law detection and sampling.

Each symbol is observed over M subslots of duration $\Delta_t = T_s/M$, with centers $t_{k,m} = kT_s + (m+1/2)\Delta_t$ and intervals $\mathcal{I}_{k,m} = [kT_s + m\Delta_t, kT_s + (m+1)\Delta_t)$. Direct square-law detection and integration are applied to the RF waveform in (7). For compactness, denote its composite complex envelope by $u_i(t) \triangleq [\mathbf{H}\mathbf{x}(t)]_i + b_i$. Then

$$\begin{aligned} z_i[k, m] &= \frac{1}{\Delta_t} \int_{\mathcal{I}_{k,m}} [r_i^{\text{RF}}(\tau)]^2 d\tau + w_i[k, m] \\ &= \frac{1}{\Delta_t} \int_{\mathcal{I}_{k,m}} |u_i(\tau)|^2 d\tau \\ &\quad + \frac{1}{\Delta_t} \text{Re}\left\{ \int_{\mathcal{I}_{k,m}} u_i^2(\tau) e^{j4\pi f_c \tau} d\tau \right\} + w_i[k, m] \\ &\simeq \frac{1}{\Delta_t} \int_{\mathcal{I}_{k,m}} |u_i(\tau)|^2 d\tau + w_i[k, m]. \end{aligned} \quad (8)$$

Here, $w_i[k, m]$ models the effective post-detection noise. The second equality follows from $[\sqrt{2} \text{Re}\{u_i(\tau) e^{j2\pi f_c \tau}\}]^2 = |u_i(\tau)|^2 + \text{Re}\{u_i^2(\tau) e^{j4\pi f_c \tau}\}$. Because $u_i(t)$ varies on the envelope time scale, the oscillatory $2f_c$ term averages out when $\Delta_t f_c \gg 1$, yielding the last line. Importantly, the retained envelope energy satisfies $|u_i(t)|^2 = |[\mathbf{H}\mathbf{x}(t)]_i|^2 + |b_i|^2 + 2 \text{Re}\{b_i^* [\mathbf{H}\mathbf{x}(t)]_i\}$; hence, integration removes the carrier-frequency oscillation but preserves the array and symbol phases through their interference with the known reference. Equation (8) is therefore an analytical representation of the measured RF energy, not coherent downconversion or I/Q sampling. If the envelope is approximately constant within a subslot, stacking the N_r element outputs yields

$$\mathbf{z}[k, m] = |\mathbf{H}\mathbf{x}[k, m] + \mathbf{b}|^2 + \mathbf{w}[k, m] \in \mathbb{R}^{N_r}, \quad (9)$$

where $\mathbf{x}[k, m] = \mathbf{x}(t_{k,m})$ and the squared modulus is element-wise. The MN_r samples associated with each symbol are retained as a vector because their spatial and temporal variations provide the diversity needed for recovery.

To relate these energy samples to the transmitted symbols, define $\mathbf{D}_P = \text{diag}(\sqrt{P_1}, \dots, \sqrt{P_L})$, $[\mathbf{P}]_{kM+m,j} = p(t_{k,m} - jT_s)$, and let $\mathbf{q}$ collect the corresponding samples of $q(t)$. Stack $\mathbf{s} = [\mathbf{s}_1^T, \dots, \mathbf{s}_L^T]^T$ and set $\boldsymbol{\xi} = [\xi_1, \dots, \xi_L]^T$. The matrix $\mathbf{P}_{\text{tx}} \triangleq \mathbf{D}_P \otimes \mathbf{P} \in \mathbb{C}^{LKM \times LK}$ maps $\mathbf{s}$ to the sampled user envelopes, while $\mathbf{x}_0 \triangleq (\mathbf{D}_P \boldsymbol{\xi}) \otimes \mathbf{q}$ represents the deterministic transmit-side component; here, $\otimes$ denotes the Kronecker product.

For general pulse shaping, neighboring symbols may overlap, so the KMN_r measurements must be processed jointly. Let $\boldsymbol{\Pi}$ reorder the user-major envelope $\mathbf{P}_{\text{tx}}\mathbf{s} + \mathbf{x}_0$ into time-major form. Stacking (9) over all subslots then yields the compact block-level model

$$\mathbf{z} = |\mathbf{G}\mathbf{s} + \mathbf{b}_{\text{eq}}|^2 + \mathbf{w}, \quad (10)$$

where $\mathbf{G} \triangleq (\mathbf{I}_{KM} \otimes \mathbf{H}) \boldsymbol{\Pi} \mathbf{P}_{\text{tx}} \in \mathbb{C}^{KMN_r \times LK}$ and $\mathbf{b}_{\text{eq}} \triangleq \mathbf{1}_{KM} \otimes \mathbf{b} + (\mathbf{I}_{KM} \otimes \mathbf{H}) \boldsymbol{\Pi} \mathbf{x}_0$. Thus, $\mathbf{b}_{\text{eq}}$ combines the receiver reference with the deterministic transmit-side contribution, while $\mathbf{s} \in \mathcal{S}^{LK}$ and $\mathbf{z}, \mathbf{w} \in \mathbb{R}^{KMN_r}$. When intersymbol overlap is negligible, (10) separates into symbol-wise models; otherwise, it describes joint block detection.

D. Two-Phase Transmission Protocol and Unified Observation Model

The receiver uses a training phase to estimate the channel from known pilots and a data phase to detect finite-alphabet symbols using the estimated channel. Both phases follow (10), but their unknowns have different structures.

1) *Training Phase*: Let $\mathbf{x}_p[t] \in \mathbb{C}^L$ be the known waveform in pilot subslot $t = 0, \dots, T_p - 1$, and define $\mathbf{h} = \text{vec}(\mathbf{H})$. Stacking the element-wise observations and using $\mathbf{H}\mathbf{x}_p[t] = (\mathbf{x}_p^T[t] \otimes \mathbf{I}_{N_r}) \mathbf{h}$ gives

$$\mathbf{z}_p = |\mathbf{G}_p \mathbf{h} + \mathbf{b}_p|^2 + \mathbf{w}_p, \quad (11)$$

where $\mathbf{G}_p$ stacks $\mathbf{x}_p^T[t] \otimes \mathbf{I}_{N_r}$, and $\mathbf{b}_p$ and $\mathbf{w}_p$ stack the reference and noise. The far-field structure in (5) will later reduce the unknown from the full vector $\mathbf{h}$ to the DOAs and complex path gains.

2) *Data Transmission Phase*: For a data block of K_d symbols per user, let $\mathbf{s}_d \in \mathcal{S}^{LK_d}$ denote the stacked symbol vector. Replacing $\mathbf{H}$ in (10) by its estimate $\hat{\mathbf{H}}$ yields

$$\mathbf{z}_d = |\mathbf{G}_d(\hat{\mathbf{H}}) \mathbf{s}_d + \mathbf{b}_d|^2 + \mathbf{w}_d, \quad (12)$$

where $Q_d = N_r M K_d$ and the known offset $\mathbf{b}_d$ includes the receiver reference and any deterministic transmit-side component. Thus, (11) and (12) have the same affine intensity structure, while the former contains continuous channel parameters and the latter a finite-alphabet vector.

III. UNIQUENESS AND STABILITY OF REFERENCE-ASSISTED ENERGY DETECTION

Reliable recovery from intensity-only measurements requires more than the existence of a solution: distinct channels or symbol vectors must produce distinct energy observations, and small measurement perturbations must not cause arbitrarily large reconstruction errors. These requirements are captured by uniqueness and stability, respectively, and are fundamental to determining whether the reference wave and RIS sampling provide sufficient information. Exploiting the quadratic structure of the reference-assisted energy mapping, we establish an exact secant-based uniqueness condition and derive lower and upper Lipschitz bounds that quantify robustness. The resulting framework applies to both continuous channel estimation and finite-alphabet symbol detection.

A. Unified Affine Phase-Retrieval Model

The training- and data-phase observations derived in Section II share the form

$$\mathbf{z} = \mathcal{T}_{\mathbf{G}, \mathbf{b}}(\boldsymbol{\theta}) + \mathbf{w}, \quad \mathcal{T}_{\mathbf{G}, \mathbf{b}}(\boldsymbol{\theta}) \triangleq |\mathbf{G}\boldsymbol{\theta} + \mathbf{b}|^2, \quad (13)$$

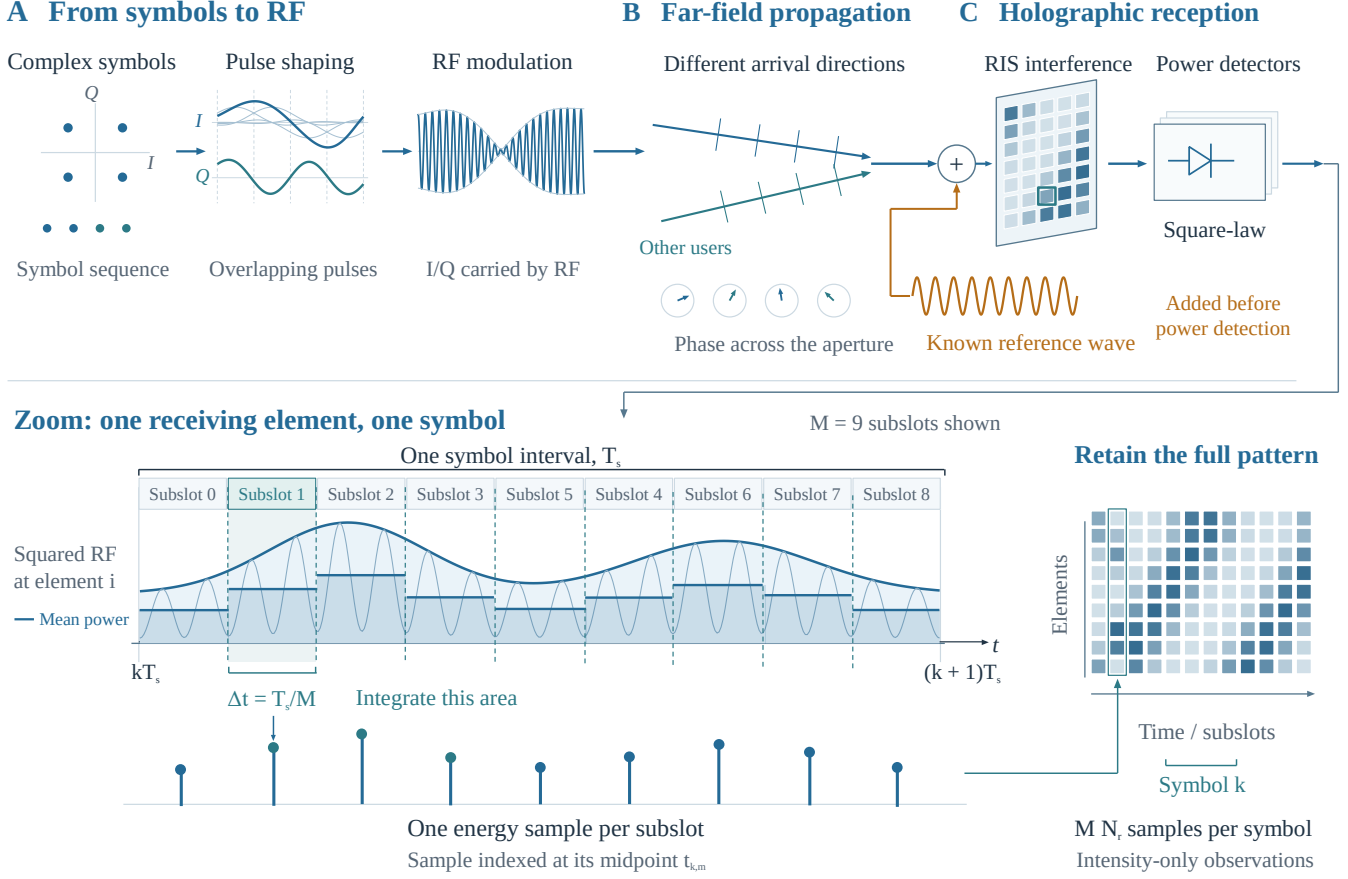


Fig. 2. Reference-assisted holographic reception model. Finite-alphabet symbols are pulse-shaped and upconverted to RF signals. After far-field propagation, the incident fields from different users form direction-dependent phase patterns across the holographic aperture. A known reference wave is added at the RF front end before square-law detection, converting relative phase information into intensity variations. Each symbol interval is divided into M subslots, and the receiver preserves the resulting element-subslot intensity pattern for channel estimation and symbol detection.

where Q is the total number of scalar intensity observations, D is the complex dimension of the unknown vector, $\mathbf{G} \in \mathbb{C}^{Q \times D}$, $\boldsymbol{\theta} \in \mathcal{X} \subseteq \mathbb{C}^D$, $\mathbf{b} \in \mathbb{C}^Q$, and $\mathbf{z}, \mathbf{w} \in \mathbb{R}^Q$. Here, $\mathcal{X}$ denotes the feasible set of the unknown parameter, and the modulus and square are applied element-wise. This is an affine phase-retrieval model [40]. For full-channel estimation in the training phase, $\boldsymbol{\theta} = \mathbf{h} = \text{vec}(\mathbf{H})$, $Q = N_r T_p$, and $D = N_r L$. If the channel is represented by a lower-dimensional linear complex parameter, D is replaced by the dimension of that parameter. In the data phase, $\boldsymbol{\theta} = \mathbf{s}_d \in \mathcal{S}^{LK_d}$, $Q = Q_d$, and $D = LK_d$.

The reference changes the ambiguity class fundamentally. If $\mathbf{b} = 0$, $\mathcal{T}_{\mathbf{G},0}(e^{j\varphi}\boldsymbol{\theta}) = \mathcal{T}_{\mathbf{G},0}(\boldsymbol{\theta})$, so uniqueness is defined only modulo a global phase. If $\mathbf{b} \neq 0$, the expansion

$$\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}) = |\mathbf{G}\boldsymbol{\theta}|^2 + |\mathbf{b}|^2 + 2 \text{Re}\{\mathbf{b}^* \odot \mathbf{G}\boldsymbol{\theta}\} \quad (14)$$

contains phase-sensitive interference, so absolute uniqueness on $\mathcal{X}$ becomes possible. A nonzero reference is not sufficient by itself, however: its phases must probe every feasible field difference in nondegenerate real directions.

For $\boldsymbol{\theta} \in \mathbb{C}^D$, define the realification $\boldsymbol{\theta}_{\mathbb{R}} = [\text{Re}\{\boldsymbol{\theta}\}^T, \text{Im}\{\boldsymbol{\theta}\}^T]^T \in \mathbb{R}^{2D}$. For two distinct parameters, define the energy-domain distance

$$d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2) \triangleq \|\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_1) - \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_2)\|_2. \quad (15)$$

The optimal lower and upper stability bounds on $\mathcal{X}$ are

$$\alpha_{\mathcal{X}} \triangleq \inf_{\substack{\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X} \\ \boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2}} \frac{d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2)}{\|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2}, \quad (16)$$

$$\beta_{\mathcal{X}} \triangleq \sup_{\substack{\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X} \\ \boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2}} \frac{d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2)}{\|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2}. \quad (17)$$

Thus, the inverse problem is globally stable on $\mathcal{X}$ if $0 < \alpha_{\mathcal{X}} \leq \beta_{\mathcal{X}} < \infty$. The lower bound is the important one: it excludes pairs that are far apart in parameter space but arbitrarily close in energy space.

B. Exact Secant Identity and Global Uniqueness

The quadratic structure of (13) gives an exact identity, which is stronger than a first-order approximation. For a point $\boldsymbol{\vartheta} \in \mathbb{C}^D$, define the real Jacobian

$$\mathbf{J}(\boldsymbol{\vartheta}) \triangleq 2 [\text{Re}\{\text{diag}[(\mathbf{G}\boldsymbol{\vartheta} + \mathbf{b})^*] \mathbf{G}\}, \\ - \text{Im}\{\text{diag}[(\mathbf{G}\boldsymbol{\vartheta} + \mathbf{b})^*] \mathbf{G}\}] \in \mathbb{R}^{Q \times 2D}. \quad (18)$$

Lemma 1 (Exact midpoint factorization). *For any $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathbb{C}^D$, let $\boldsymbol{\mu} = (\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2)/2$ and $\boldsymbol{\delta} = \boldsymbol{\theta}_1 - \boldsymbol{\theta}_2$. Then*

$$\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_1) - \mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta}_2) = \mathbf{J}(\boldsymbol{\mu})\boldsymbol{\delta}_{\mathbb{R}}. \quad (19)$$

Proof. Applying $|a|^2 - |c|^2 = \text{Re}\{(a+c)^*(a-c)\}$ element-wise gives $\text{Re}\{[\mathbf{G}(\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2) + 2\mathbf{b}]^* \odot \mathbf{G}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)\}$, which equals (19) after realification. $\square$

Theorem 1 (Pairwise uniqueness criterion). *The mapping $\mathcal{T}_{\mathbf{G},\mathbf{b}}$ is injective on $\mathcal{X}$ if and only if, for every distinct $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X}$,*

$$\mathbf{J}\left(\frac{\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2}{2}\right)(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)_{\mathbb{R}} \neq \mathbf{0}. \quad (20)$$

Proof. The claim follows directly from Lemma 1. $\square$

For a full-dimensional continuous set, $\text{rank}(\mathbf{G}) = D$ is necessary but not sufficient: a nonzero field difference can still vanish after the reference-dependent real projection. A sharper conclusion is available for finite hypotheses.

Corollary 1 (Generic-reference uniqueness for finite hypotheses). *Let $\Xi = \{\xi_1, \dots, \xi_M\}$ be finite, with pre-reference field codewords $\mathbf{u}_i = \mathbf{f}(\xi_i) \in \mathbb{C}^Q$. For $i \neq j$, set $\mathbf{d}_{ij} = \mathbf{u}_i - \mathbf{u}_j$ and $\mathbf{a}_{ij} = (|\mathbf{u}_i|^2 - |\mathbf{u}_j|^2)/2$. A pair with $\mathbf{d}_{ij} = \mathbf{0}$ is indistinguishable for every reference; otherwise, its ambiguous references form the proper real affine set*

$$\mathcal{B}_{ij}^{\text{bad}} = \{\mathbf{b} \in \mathbb{C}^Q : \text{Re}\{\mathbf{b}^* \odot \mathbf{d}_{ij}\} = -\mathbf{a}_{ij}\}. \quad (21)$$

If all field codewords are distinct, the finite union $\bigcup_{i < j} \mathcal{B}_{ij}^{\text{bad}}$ has measure zero in $\mathbb{C}^Q$; hence, an absolutely continuously distributed reference makes $|\mathbf{u}_i + \mathbf{b}|^2$ pairwise distinct with probability one.

Proof. Expanding the energy difference gives $2\mathbf{a}_{ij} + 2\text{Re}\{\mathbf{b}^* \odot \mathbf{d}_{ij}\}$. For $d_{ij,q} \neq 0$, the ambiguous values are $b_q = (-a_{ij,q}/|d_{ij,q}|^2 + j\tau_q)d_{ij,q}$, $\tau_q \in \mathbb{R}$, which impose one real affine constraint per active coordinate. Thus, each $\mathcal{B}_{ij}^{\text{bad}}$ is proper, and their finite union has measure zero. $\square$

For $b_q = \rho_q e^{j\chi_q}$, an active coordinate is ambiguous only when $\rho_q |d_{ij,q}| \cos(\angle d_{ij,q} - \chi_q) = -a_{ij,q}$, which excludes at most two phases. Continuous phases are therefore ambiguity-free almost surely when the reference support meets every field difference; quantized phases can be checked directly. For $\mathbf{f}(\boldsymbol{\theta}) = \mathbf{G}\boldsymbol{\theta}$, the premise reduces to $\mathbf{G}(\boldsymbol{\theta}_i - \boldsymbol{\theta}_j) \neq \mathbf{0}$. Full column rank is sufficient but unnecessary on a finite set, while a difference already annihilated by $\mathbf{G}$ cannot be restored by any reference. The result also applies to $\mathbf{u}_i = \mathbf{g}_p(\boldsymbol{\eta}_i)$ on a finite angular grid when its nuisance parameters are fixed or discretized; continuous gains require the parametric analysis below.

Generic injectivity on a finite set does not by itself ensure a useful noise margin. Returning to a general feasible set, let $\mathcal{M}_{\mathcal{X}} = \{(\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2)/2 : \boldsymbol{\theta}_1, \boldsymbol{\theta}_2 \in \mathcal{X}\}$ be the midpoint set. The exact identity yields computable two-sided bounds.

Theorem 2 (Jacobian stability bounds). *Define*

$$\underline{\sigma}_J \triangleq \inf_{\boldsymbol{\mu} \in \mathcal{M}_{\mathcal{X}}} \sigma_{\min}(\mathbf{J}(\boldsymbol{\mu})), \quad \bar{\sigma}_J \triangleq \sup_{\boldsymbol{\mu} \in \mathcal{M}_{\mathcal{X}}} \sigma_{\max}(\mathbf{J}(\boldsymbol{\mu})). \quad (22)$$

Then

$$\underline{\sigma}_J \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2 \leq d_E(\boldsymbol{\theta}_1, \boldsymbol{\theta}_2) \leq \bar{\sigma}_J \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|_2 \quad (23)$$

for all pairs in $\mathcal{X}$. Consequently, $\alpha_{\mathcal{X}} \geq \underline{\sigma}_J$ and $\beta_{\mathcal{X}} \leq \bar{\sigma}_J$.

Proof. Apply the singular-value inequalities to (19) over the midpoint set. $\square$

The bound $\underline{\sigma}_J > 0$ is a convenient sufficient condition for global stability, but it can be conservative because it tests every direction at every midpoint, including directions that may not connect two feasible points. The exact constant in (16), or equivalently the restricted secants in (19), is the sharp criterion.

The stability constants also depend on the balance between the received field and the reference wave. To make this dependence explicit, let $u_q(\boldsymbol{\theta}) = [\mathbf{G}\boldsymbol{\theta}]_q$ and, whenever $u_q(\boldsymbol{\theta}) \neq 0$, define the per-measurement reference-to-signal energy ratio and the corresponding normalized interference depth as

$$\gamma_q(\boldsymbol{\theta}) \triangleq \frac{|b_q|^2}{|u_q(\boldsymbol{\theta})|^2}, \quad \kappa_q(\boldsymbol{\theta}) \triangleq \frac{2\sqrt{\gamma_q(\boldsymbol{\theta})}}{1 + \gamma_q(\boldsymbol{\theta})}. \quad (24)$$

Writing $\Delta\phi_q = \angle u_q - \angle b_q$, the q th noiseless measurement becomes

$$[\mathcal{T}_{\mathbf{G},\mathbf{b}}(\boldsymbol{\theta})]_q = (|u_q|^2 + |b_q|^2)[1 + \kappa_q(\boldsymbol{\theta}) \cos(\Delta\phi_q)]. \quad (25)$$

Here $0 \leq \kappa_q \leq 1$, with its maximum attained when the two incident fields have equal energy. It approaches zero when either field dominates, showing that the phase-sensitive interference then occupies only a small fraction of the detected energy.

More directly, consider the global-phase perturbation $e^{j\varphi}\boldsymbol{\theta}$. Its tangent direction satisfies

$$\begin{aligned} \mathbf{J}(\boldsymbol{\theta})(j\boldsymbol{\theta})_{\mathbb{R}} &= -2\text{Im}\{\mathbf{b}^* \odot \mathbf{G}\boldsymbol{\theta}\}, \\ \sigma_{\min}(\mathbf{J}(\boldsymbol{\theta})) &\leq \frac{2\|\text{Im}\{\mathbf{b}^* \odot \mathbf{G}\boldsymbol{\theta}\}\|_2}{\|\boldsymbol{\theta}\|_2}. \end{aligned} \quad (26)$$

Thus, as the reference energy vanishes relative to a fixed nonzero received field, the global-phase direction becomes unobservable and the lower stability margin collapses to that of homogeneous phase retrieval. Energy balance alone is not sufficient, however: even for $\gamma_q = 1$, the contribution in (26) vanishes when the two fields are locally phase aligned. Reference-phase diversity is therefore required to prevent all measurements from becoming insensitive to the same parameter direction.

Equation (24) characterizes relative phase contrast, whereas the absolute Lipschitz constants also depend on the total field level, $\mathbf{G}$, and the reference phases. In particular, under the fixed-variance post-detection noise model used here, increasing the reference amplitude can increase the absolute Jacobian magnitude even when the normalized interference depth decreases; a finite optimal energy ratio would additionally depend on detector dynamic range or power-dependent noise. Consequently, the reference-to-signal energy ratio is a key stability variable but should be assessed jointly with $\underline{\sigma}_J$ rather than used as a stand-alone criterion.

Finally, affine phase retrievability guarantees two-sided Lipschitz bounds on a compact feasible set [40, Theorem 4.1], whereas no uniform bi-Lipschitz constants exist over the entire unbounded space [40, Proposition 4.1]. The bounded channel-gain and transmit-power regions encountered here therefore provide the appropriate domain for Proposition 2. Within this domain, (24)–(26) explain how reference strength and phase diversity jointly control the stability margin.

C. Parametric Channel Uniqueness Requirements

The affine phase-retrieval result in [40, Theorem 3.1] states that a mapping on $\mathbb{C}^D$ is globally injective if and only if its real Jacobian has rank $2D$ at every point. For unstructured channel training, $D_h = N_r L$, $Q_p = N_r T_p$, and the unknown is $\mathbf{h} = \text{vec}(\mathbf{H}) \in \mathbb{C}^{D_h}$. The universal minimum-measurement result in [40, Theorem 3.2] therefore gives the necessary condition

$$Q_p = N_r T_p \geq 3D_h = 3N_r L \implies T_p \geq 3L. \quad (27)$$

This bound concerns global recovery over the entire space $\mathbb{C}^{N_r L}$; hence, increasing N_r does not reduce the required number of pilot subslots because it increases the number of unknown channel coefficients by the same factor. A generic unconstrained affine measurement pair is sufficient when $Q_p \geq 4D_h - 1$ [40, Theorem 3.4]. This generic result is only a benchmark here, however, because $\mathbf{G}_p = \mathbf{X}_p \otimes \mathbf{I}_{N_r}$ has a structured Kronecker form.

The far-field model reduces the channel to a real parameter manifold. Let d_a be the number of angular parameters per user and define $p = (d_a + 2)L$, where the additional two parameters are the real and imaginary parts of each path gain. With $\mathbf{h}_{\mathbb{R}}(\boldsymbol{\eta}) \triangleq [\text{Re}\{\mathbf{h}(\boldsymbol{\eta})\}^T, \text{Im}\{\mathbf{h}(\boldsymbol{\eta})\}^T]^T$ and $\mathbf{H}_{\boldsymbol{\eta}} \triangleq \partial \mathbf{h}_{\mathbb{R}} / \partial \boldsymbol{\eta}^T$, the chain rule gives the parameter-domain Jacobian

$$\mathbf{J}_{\boldsymbol{\eta}}(\boldsymbol{\eta}) = \mathbf{J}_h(\mathbf{h}(\boldsymbol{\eta})) \mathbf{H}_{\boldsymbol{\eta}}(\boldsymbol{\eta}) \in \mathbb{R}^{Q_p \times p}, \quad (28)$$

where $\mathbf{J}_h$ is (18) evaluated with $\mathbf{G} = \mathbf{G}_p$ and $\mathbf{b} = \mathbf{b}_p$. If each of the K_p pilot symbols is observed over M subslots, then $T_p = MK_p$.

Proposition 1 (Parametric training count). *At an interior regular point, a differentiable local inverse requires*

$$\text{rank}(\mathbf{J}_{\boldsymbol{\eta}}(\boldsymbol{\eta})) = p. \quad (29)$$

Conversely, this rank condition makes the training map a local embedding. Consequently, a necessary measurement count is

$$\begin{aligned} N_r T_p = N_r M K_p &\geq (d_a + 2)L, \\ K_p &\geq \left\lceil \frac{(d_a + 2)L}{M N_r} \right\rceil. \end{aligned} \quad (30)$$

Thus, the right-hand side is $4L$ for the planar-array model with two angles per user, $3L$ for a linear-array model with one angle, and $2L$ when the DOAs are known and only the complex gains are estimated.

Proof. A differentiable local inverse requires an injective differential, whereas full column rank in (29) makes the training map a local embedding by the constant-rank theorem. Since $\text{rank}(\mathbf{J}_{\boldsymbol{\eta}}) \leq \min\{N_r M K_p, p\}$, the stated counts follow from $p = (d_a + 2)L$. $\square$

The count in (30) is necessary but not sufficient. Full rank additionally requires nonzero path gains, a two-dimensional array aperture for two-dimensional DOA estimation, an array geometry without angular aliasing, pilots that preserve all tangent directions, and reference phases that do not annihilate those directions after the real projection in (18). More RIS elements can therefore trade for fewer temporal measurements only under the far-field parameterization, whose dimension remains $4L$. Finally, (29) is a local condition; global parameter

uniqueness also requires the pairwise condition (20) on the channel manifold, together with an ambiguity-free angular region and, for unlabeled paths, a fixed ordering convention.

Within an identifiable neighborhood, the smallest singular value gives the local inverse sensitivity. With $\mathcal{T}_p(\boldsymbol{\eta}) \triangleq |\mathbf{G}_p \mathbf{h}(\boldsymbol{\eta}) + \mathbf{b}_p|^2$,

$$\liminf_{\|\boldsymbol{\delta}\|_2 \rightarrow 0} \frac{\|\mathcal{T}_p(\boldsymbol{\eta} + \boldsymbol{\delta}) - \mathcal{T}_p(\boldsymbol{\eta})\|_2}{\|\boldsymbol{\delta}\|_2} = \sigma_{\min}(\mathbf{J}_{\boldsymbol{\eta}}(\boldsymbol{\eta})). \quad (31)$$

Thus, the rank condition establishes local uniqueness, whereas the magnitude of $\sigma_{\min}(\mathbf{J}_{\boldsymbol{\eta}})$ determines how strongly noise is amplified.

For the unstructured model, $\mathbf{G}_p = \mathbf{X}_p \otimes \mathbf{I}_{N_r}$, so $\text{rank}(\mathbf{G}_p) = N_r \text{rank}(\mathbf{X}_p)$ and $\sigma_{\min}(\mathbf{G}_p) = \sigma_{\min}(\mathbf{X}_p)$. The resulting necessary hierarchy is

$$\begin{aligned} T_p &\geq L && \text{for complex-field preservation,} \\ T_p &\geq 2L && \text{for regular local intensity recovery,} \\ T_p &\geq 3L && \text{for global unstructured recovery.} \end{aligned} \quad (32)$$

These counts do not replace the pilot- and reference-diversity requirements. When the DOAs are known, only the $2L$ real gain parameters remain, giving $N_r T_p \geq 2L$; nevertheless, the stacked matrix $\text{col}_t\{\mathbf{V} \text{diag}(\mathbf{x}_p[t])\}$ must preserve every gain direction. Closely spaced angles reduce $\sigma_{\min}(\mathbf{V})$ and hence both the field- and energy-domain stability margins.

D. Finite-Alphabet Uniqueness and Noise Margin

For data detection, let $\mathcal{X}_d = \mathcal{S}^{LK_d}$. This specializes Corollary 1 to $\mathbf{u}_i = \mathbf{G}_d \mathbf{s}_i$. Since the candidate set is finite, noiseless uniqueness is equivalent to a positive minimum separation. Define

$$d_{\min, d}^{(E)} \triangleq \min_{\substack{\mathbf{s}_1, \mathbf{s}_2 \in \mathcal{X}_d \\ \mathbf{s}_1 \neq \mathbf{s}_2}} \|\mathbf{G}_d \mathbf{s}_1 + \mathbf{b}_d\|^2 - \|\mathbf{G}_d \mathbf{s}_2 + \mathbf{b}_d\|^2. \quad (33)$$

Corollary 2 (Noise-limited finite-alphabet reliability). *For an arbitrary additive perturbation, the nearest-intensity detector returns the transmitted symbol vector whenever*

$$\|\mathbf{w}\|_2 < \frac{d_{\min, d}^{(E)}}{2}. \quad (34)$$

Under exact CSI and additive white Gaussian post-detection noise $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \sigma_w^2 \mathbf{I})$, the error probability of the binary ML comparison between distinct candidates $\mathbf{s}_i$ and $\mathbf{s}_j$ is

$$P_{i \rightarrow j} = Q\left(\frac{d_E(\mathbf{s}_i, \mathbf{s}_j)}{2\sigma_w}\right). \quad (35)$$

Consequently, because the minimum is attained on the finite candidate set,

$$P_{\text{pair, worst}} \triangleq \max_{i \neq j} P_{i \rightarrow j} = Q\left(\frac{d_{\min, d}^{(E)}}{2\sigma_w}\right). \quad (36)$$

If $M_c = |\mathcal{X}_d| = |\mathcal{S}|^{LK_d}$ is the number of candidate symbol vectors, the union bound gives

$$P_B \leq (M_c - 1) Q\left(\frac{d_{\min, d}^{(E)}}{2\sigma_w}\right). \quad (37)$$

Using $Q(x) \leq \frac{1}{2}e^{-x^2/2}$ further yields

$$P_B \leq \frac{M_c - 1}{2} \exp \left[-\frac{(d_{\min,d}^{(E)})^2}{8\sigma_w^2} \right]. \quad (38)$$

Proof. Let $\mu_k = |\mathbf{G}_d \mathbf{s}_k + \mathbf{b}_d|^2$. The triangle inequality gives the decoding radius in (34). If $\mathbf{s}_i$ is transmitted, the binary ML comparison favors $\mathbf{s}_j$ only if $\mathbf{w}^T(\mu_j - \mu_i) \geq d_E^2(\mathbf{s}_i, \mathbf{s}_j)/2$. The left-hand side is Gaussian with variance $\sigma_w^2 d_E^2(\mathbf{s}_i, \mathbf{s}_j)$, which gives (35). Taking the closest pair, applying the union bound over the remaining $M_c - 1$ candidates, and using the standard Gaussian-tail inequality yield the final three bounds. $\square$

Corollary 3 (Near-balanced reference-to-signal energy allocation). *Under the noise model of Corollary 2, suppose that the candidate fields are $\mathbf{u}_i = \mathbf{G}_d \mathbf{s}_i$. Define the average received-field energy and reference energy per measurement as*

$$P_s \triangleq \frac{1}{M_c Q_d} \sum_{i=1}^{M_c} \|\mathbf{u}_i\|_2^2, \quad P_b \triangleq \frac{1}{Q_d} \|\mathbf{b}_d\|_2^2. \quad (39)$$

And the total energy is $P_T = P_s + P_b$. Writing $\mathbf{u}_i = \sqrt{P_s} \mathbf{v}_i$ gives $(M_c Q_d)^{-1} \sum_i \|\mathbf{v}_i\|_2^2 = 1$. Suppose that P_T is fixed and four references $\sqrt{P_b} e^{j\phi_k}$ are used, where $\phi_k \in \{0, \pi/2, \pi, 3\pi/2\}$. Then

$$\begin{aligned} d_{E,ij}^2 &= 4P_s^2 A_{ij} + 8P_s P_b B_{ij}, \\ A_{ij} &= \left| \|\mathbf{v}_i\|^2 - \|\mathbf{v}_j\|^2 \right|_2^2, \\ B_{ij} &= \|\mathbf{v}_i - \mathbf{v}_j\|_2^2. \end{aligned} \quad (40)$$

For $0 \leq A_{ij} < B_{ij}$, this distance is maximized by

$$P_s^* = \frac{B_{ij}}{2B_{ij} - A_{ij}} P_T, \quad \frac{P_b^*}{P_s^*} = 1 - \frac{A_{ij}}{B_{ij}}. \quad (41)$$

In particular, phase-only pairs satisfy $A_{ij} = 0$ and are optimally separated by $P_s = P_b = P_T/2$. Therefore, when the minimum-distance pairs are phase dominated, $A_{ij} \ll B_{ij}$, the dominant pairwise error probabilities and the high-SNR union bound are minimized near $10 \log_{10}(P_b/P_s) = 0$ dB.

Proof. For measurement coordinate q and reference phase ϕ_k , let $a_q = |v_{i,q}|^2 - |v_{j,q}|^2$ and $\delta v_q = v_{i,q} - v_{j,q}$. The noiseless intensity difference is

$$\Delta\mu_{q,k} = P_s a_q + 2\sqrt{P_s P_b} \operatorname{Re}\{e^{-j\phi_k} \delta v_q\}. \quad (42)$$

For the four phases, the real projections sum to zero and their squared sum is $2|\delta v_q|^2$. Summing $|\Delta\mu_{q,k}|^2$ over q and k therefore gives (40). Substituting $P_b = P_T - P_s$ and differentiating yields (41); its second derivative is negative for $A_{ij} < B_{ij}$. When $A_{ij} = 0$,

$$d_{E,ij}^2 = 8B_{ij} P_s (P_T - P_s) \leq 2B_{ij} P_T^2, \quad (43)$$

with equality at equal energy. Since the Gaussian tail in (35) decreases monotonically with $d_{E,ij}$, maximizing the relevant pairwise distances minimizes their error probabilities and the associated union-bound terms. $\square$

The finite optimum in Corollary 3 relies on a fixed total-energy budget. If P_s and σ_w^2 are fixed while only the reference power is increased, the ideal model does not in general predict an optimum at equal energy.

E. A Constructive Four-Phase Reference Bound

The preceding conditions are exact but depend on the unknown midpoint. A phase-complete reference pattern gives a simple midpoint-independent sufficient condition. Suppose a base field sample $u_q = [\mathbf{G}\boldsymbol{\theta}]_q$ is repeated with four known references

$$b_{q,k} = \rho_q e^{j\phi_k}, \quad \phi_k \in \{0, \pi/2, \pi, 3\pi/2\}, \quad \rho_q > 0. \quad (44)$$

Denote the resulting stacked mapping by $\mathcal{T}_{4\text{ph}}(\boldsymbol{\theta})$.

Theorem 3 (Four-phase global lower bound). *For any $\boldsymbol{\theta}_1, \boldsymbol{\theta}_2$,*

$$\begin{aligned} \|\mathcal{T}_{4\text{ph}}(\boldsymbol{\theta}_1) - \mathcal{T}_{4\text{ph}}(\boldsymbol{\theta}_2)\|_2^2 \\ = 4\| |\mathbf{G}\boldsymbol{\theta}_1|^2 - |\mathbf{G}\boldsymbol{\theta}_2|^2 \|_2^2 + 8 \sum_q \rho_q^2 \|\mathbf{G}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)\|_q^2. \end{aligned} \quad (45)$$

Consequently, with $\rho_{\min} = \min_q \rho_q$,

$$\|\mathcal{T}_{4\text{ph}}(\boldsymbol{\theta}_1) - \mathcal{T}_{4\text{ph}}(\boldsymbol{\theta}_2)\|_2 \geq 2\sqrt{2}\rho_{\min} \|\mathbf{G}(\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2)\|_2. \quad (46)$$

If $\mathbf{G}$ has full column rank, then

$$\alpha_{\mathcal{X}} \geq 2\sqrt{2}\rho_{\min} \sigma_{\min}(\mathbf{G}) > 0 \quad (47)$$

for every feasible set $\mathcal{X}$. For a finite alphabet, full column rank can be replaced by $\text{Null}(\mathbf{G}) \cap \mathcal{D}_{\mathcal{X}} = \emptyset$.

Proof. For one field coordinate, put $c_q = |u_{1,q}|^2 - |u_{2,q}|^2$ and $\Delta u_q = u_{1,q} - u_{2,q}$. Summing $[c_q + 2\rho_q \operatorname{Re}\{e^{-j\phi_k} \Delta u_q\}]^2$ over the four phases cancels all cross terms and gives $4c_q^2 + 8\rho_q^2 |\Delta u_q|^2$. Summing over q proves (45); the remaining claims follow from singular value or restricted-difference bounds. $\square$

The construction also has a transparent holographic interpretation. Opposite reference phases cancel the unknown self-intensity:

$$\operatorname{Re}\{u_q\} = \frac{z_{q,0} - z_{q,\pi}}{4\rho_q}, \quad \operatorname{Im}\{u_q\} = \frac{z_{q,\pi/2} - z_{q,3\pi/2}}{4\rho_q}. \quad (48)$$

Thus, four phase-shifted energy measurements recover two orthogonal real projections of each complex field sample. This proves explicitly that reference phase diversity can remove the global phase ambiguity and yields the stability lower bound in (47). Fewer or hardware-constrained phases may still be unique, but their quality should be evaluated through $\alpha_{\mathcal{X}}$, $\underline{\sigma}_J$, or $d_{\min,d}^{(E)}$, rather than through the rank of $\mathbf{G}$ alone.

The resulting design criteria are therefore

$$\max_{\mathbf{b}_p \in \mathcal{B}_p} \inf_{\boldsymbol{\theta} \in \mathcal{X}_p} \sigma_{\min}[\mathbf{J}_p(\boldsymbol{\theta})] \quad \text{and} \quad \max_{\mathbf{b}_d \in \mathcal{B}_d} d_{\min,d}^{(E)} \quad (49)$$

for continuous training and discrete data detection, respectively. The first maximizes a uniform local/global lower bound, whereas the second directly maximizes the finite-alphabet decision margin.

IV. ESTIMATION AND DETECTION ALGORITHMS

Based on the uniqueness and stability analysis in Section III, this section develops practical recovery algorithms for the proposed holographic RIS receiver. The key point is that the algorithms should not be chosen only according to the algebraic rank of the equivalent matrix. Instead, the training stage should operate in a region where the intensity-domain Jacobian is well conditioned, while the data detector should preserve a positive energy-domain distance between finite-alphabet candidates.

The resulting receiver follows a two-stage structure. In the training stage, the unknown variables are the DOAs and the complex path gains. They are continuous parameters, and the far-field model provides a differentiable low-dimensional representation. A damped Gauss–Newton refinement is therefore used after a coarse initialization. In the data stage, the channel estimate is fixed and the unknown vector belongs to a finite constellation. Maximum-likelihood (ML) detection gives the finite-alphabet benchmark associated with the energy-domain minimum distance, whereas a reference-assisted Gerchberg–Saxton (GS) detector provides a lower-complexity iterative implementation for repeated data blocks.

A. Overall Two-Stage Recovery Framework

The proposed receiver retains the two-phase structure of Section II. During training, the far-field model in (5) is parameterized by the real vector

$$\boldsymbol{\eta} = [\boldsymbol{\theta}^T \quad \boldsymbol{\phi}^T \quad \text{Re}\{\boldsymbol{\alpha}\}^T \quad \text{Im}\{\boldsymbol{\alpha}\}^T]^T \in \mathbb{R}^{D_\eta}, \quad (50)$$

where $\boldsymbol{\theta}$, $\boldsymbol{\phi}$, and $\boldsymbol{\alpha}$ collect the polar angles, azimuth angles, and complex path gains, respectively; thus, $D_\eta = 4L$. Let $\mathbf{g}_p(\boldsymbol{\eta}) = \text{col}_t\{\mathbf{H}(\boldsymbol{\eta})\mathbf{x}_p[t]\} \in \mathbb{C}^{Q_p}$, where $Q_p = N_r T_p$. The parametric training model is

$$\mathbf{z}_p = |\mathbf{g}_p(\boldsymbol{\eta}) + \mathbf{b}_p|^2 + \mathbf{w}_p, \quad (51)$$

After Gauss–Newton refinement, the receiver reconstructs $\hat{\mathbf{H}} = \mathbf{H}(\hat{\boldsymbol{\eta}})$ and forms $\hat{\mathbf{G}}_d = \mathbf{G}_d(\hat{\mathbf{H}})$ in (12). The training reference is designed to improve the conditioning of $\mathbf{J}_p$, whereas the data reference is designed to enlarge the finite-alphabet energy distance. The following two subsections give the corresponding estimators.

B. Joint DOA and Channel-Gain Estimation

1) *Nonlinear Least-Squares Formulation:* Based on (51), joint DOA and channel-gain estimation is formulated as

$$\hat{\boldsymbol{\eta}} = \arg \min_{\boldsymbol{\eta} \in \Omega} F_p(\boldsymbol{\eta}), \quad F_p(\boldsymbol{\eta}) \triangleq \frac{1}{2} \left\| \mathbf{z}_p - |\mathbf{g}_p(\boldsymbol{\eta}) + \mathbf{b}_p|^2 \right\|_2^2, \quad (52)$$

where Ω is the feasible parameter region. This physical parameterization uses $4L$ real unknowns rather than the $2N_r L$ real entries of an unstructured channel. Its Jacobian also exposes weak gains and poorly separated DOAs through small singular values.

2) *Jacobian of the Parametric Intensity Mapping:* At the t -th pilot instant, the noise-free received field is given by

$$\mathbf{g}_p[t] = \sum_{l=1}^L \alpha_l x_{p,l}[t] \mathbf{v}_l, \quad (53)$$

where $x_{p,l}[t]$ denotes the pilot symbol transmitted by the l -th user, corresponding to the l -th entry of the pilot vector $\mathbf{x}_p[t]$.

Let $\boldsymbol{\psi}_l \in \mathbb{R}^{N_r}$ denote the phase vector whose $\iota(n, m)$ -th entry is $[\boldsymbol{\psi}_l]_{\iota(n, m)} = \psi_{n, m}^{(l)}$. Since $\mathbf{v}_l(\theta_l, \phi_l) = \exp(j\boldsymbol{\psi}_l)$, where the exponential is applied element-wise, its angular derivatives can be written as

$$\frac{\partial \mathbf{v}_l}{\partial \theta_l} = j \frac{\partial \boldsymbol{\psi}_l}{\partial \theta_l} \odot \mathbf{v}_l, \quad \frac{\partial \mathbf{v}_l}{\partial \phi_l} = j \frac{\partial \boldsymbol{\psi}_l}{\partial \phi_l} \odot \mathbf{v}_l, \quad (54)$$

where the $\iota(n, m)$ -th entries are $[\partial \boldsymbol{\psi}_l / \partial \theta_l]_{\iota(n, m)} = k_0 \cos \theta_l (y_n \cos \phi_l + z_m \sin \phi_l)$ and $[\partial \boldsymbol{\psi}_l / \partial \phi_l]_{\iota(n, m)} = k_0 \sin \theta_l (-y_n \sin \phi_l + z_m \cos \phi_l)$.

Using (53), and defining $\alpha_{l,R} = \text{Re}\{\alpha_l\}$ and $\alpha_{l,I} = \text{Im}\{\alpha_l\}$, the field derivatives at pilot instant t are

$$\begin{aligned} \frac{\partial \mathbf{g}_p[t]}{\partial \theta_l} &= \alpha_l x_{p,l}[t] \frac{\partial \mathbf{v}_l}{\partial \theta_l}, & \frac{\partial \mathbf{g}_p[t]}{\partial \phi_l} &= \alpha_l x_{p,l}[t] \frac{\partial \mathbf{v}_l}{\partial \phi_l}, \\ \frac{\partial \mathbf{g}_p[t]}{\partial \alpha_{l,R}} &= x_{p,l}[t] \mathbf{v}_l, & \frac{\partial \mathbf{g}_p[t]}{\partial \alpha_{l,I}} &= j x_{p,l}[t] \mathbf{v}_l. \end{aligned} \quad (55)$$

The derivatives over all pilot instants are stacked in the same order as $\mathbf{g}_p(\boldsymbol{\eta})$. Thus, the Jacobian of the parametric intensity mapping is

$$\mathbf{J}_p(\boldsymbol{\eta}) = 2 \text{Re} \left\{ \text{diag}(\mathbf{u}_p^*(\boldsymbol{\eta})) \frac{\partial \mathbf{g}_p(\boldsymbol{\eta})}{\partial \boldsymbol{\eta}^T} \right\} \in \mathbb{R}^{Q_p \times D_\eta}. \quad (56)$$

The same Jacobian determines both the local identifiability discussed in Section III and the numerical behavior of the Gauss–Newton iterations. In particular, a small $\sigma_{\min}(\mathbf{J}_p)$ indicates that certain perturbations in the DOAs or path gains produce only weak intensity variations, leading to both poor estimation stability and poorly conditioned Gauss–Newton updates. This is the reason for using phase-diverse references during training: the reference term changes the real projection directions in the Jacobian and can convert a nearly invisible complex-field perturbation into a measurable intensity perturbation.

3) *Damped Gauss–Newton Update:* At iteration i , define $\mathbf{u}_p^{(i)} = \mathbf{g}_p(\boldsymbol{\eta}^{(i)}) + \mathbf{b}_p$, $\mathbf{e}_p^{(i)} = \mathbf{z}_p - \left| \mathbf{u}_p^{(i)} \right|^2$, and $\mathbf{J}_p^{(i)} = \mathbf{J}_p(\boldsymbol{\eta}^{(i)})$ denotes the intensity-domain Jacobian evaluated at the current iterate.

The damped Gauss–Newton increment is then computed as

$$\Delta \boldsymbol{\eta}^{(i)} = \left[\left(\mathbf{J}_p^{(i)} \right)^T \mathbf{J}_p^{(i)} + \lambda^{(i)} \mathbf{I} \right]^{-1} \left(\mathbf{J}_p^{(i)} \right)^T \mathbf{e}_p^{(i)}, \quad (57)$$

where $\lambda^{(i)} \geq 0$ is a damping parameter introduced to improve numerical stability when the Jacobian is poorly conditioned. The parameter vector is updated as

$$\boldsymbol{\eta}_{\text{trial}}^{(i+1)} = \Pi_\Omega \left(\boldsymbol{\eta}^{(i)} + \Delta \boldsymbol{\eta}^{(i)} \right), \quad (58)$$

where $\Pi_\Omega(\cdot)$ projects the angular parameters onto the feasible region. The trial point is accepted and the damping is reduced when it lowers F_p ; otherwise, the current iterate is retained and the damping is increased.

4) *Initialization*: Because (52) is nonconvex, the complex field is first estimated coarsely and then fitted to the far-field model. With the four-phase training reference, opposite-phase differences in (48) directly give the field samples. Stacking them gives $\hat{\mathbf{g}}_p$. A coarse unstructured channel estimate is then obtained by least squares as

$$\mathbf{h}^{(0)} = \mathbf{G}_p^\dagger \hat{\mathbf{g}}_p, \quad \mathbf{H}^{(0)} = \text{unvec}(\mathbf{h}^{(0)}). \quad (59)$$

When four separate reference phases are unavailable, the reference-dominated approximation can be used to obtain

$$\mathbf{h}^{(0)} = \arg \min_{\mathbf{h}} \|\mathbf{z}_p - |\mathbf{b}_p|^2 - 2 \text{Re}\{\mathbf{b}_p^* \odot \mathbf{G}_p \mathbf{h}\}\|_2^2, \quad (60)$$

from which $\mathbf{H}^{(0)}$ is reconstructed.

The initial DOA of each user is then obtained through a coarse angular search,

$$(\theta_l^{(0)}, \phi_l^{(0)}) = \arg \max_{(\theta, \phi) \in \mathcal{G}_\Omega} \frac{|\mathbf{v}^H(\theta, \phi) \mathbf{h}_l^{(0)}|^2}{\|\mathbf{v}(\theta, \phi)\|_2^2}, \quad (61)$$

where $\mathcal{G}_\Omega$ denotes a coarse angular grid. Given the coarse DOA estimate, the corresponding complex path gain is obtained by least-squares projection onto the steering vector. These coarse parameter estimates are then used to initialize the damped Gauss–Newton iterations.

The complete joint estimation procedure is summarized in Algorithm 1.

The Gauss–Newton method is a local optimization procedure and does not guarantee convergence to the global minimizer from an arbitrary initialization. Its use is justified here by the differentiable low-dimensional channel parameterization, the availability of pilot observations, and the possibility of obtaining a coarse angular initialization before iterative refinement.

C. Finite-Alphabet Symbol Detection

After training, $\hat{\mathbf{H}}$ is used to construct $\hat{\mathbf{G}}_d$ in (12). Because $\mathbf{s}_d \in \mathcal{S}^{D_s}$ is finite-alphabet, reliability is governed by the energy-domain separation in (33), rather than by a continuous Jacobian rank condition.

1) *Maximum-Likelihood Detection*: Assuming independent Gaussian post-detection noise with identical variance, the maximum-likelihood detector is equivalent to

$$\hat{\mathbf{s}}_{\text{ML}} = \arg \min_{\mathbf{s} \in \mathcal{S}^{D_s}} \|\mathbf{z}_d - |\hat{\mathbf{G}}_d \mathbf{s} + \mathbf{b}_d|^2\|_2^2. \quad (62)$$

The corresponding estimated minimum distance is obtained from (33) by replacing $\mathbf{G}_d$ with $\hat{\mathbf{G}}_d$. A positive value ensures distinct noiseless intensity vectors, but exhaustive ML evaluates $|\mathcal{S}|^{D_s}$ candidates and is therefore used mainly as a short-block benchmark.

2) *Linearized Initialization and Candidate Screening*: If the reference-induced interference dominates the unknown self-intensity, $\mathbf{z}_d - |\mathbf{b}_d|^2 \approx \mathbf{A}_d \bar{\mathbf{s}}_d$, where $\bar{\mathbf{s}} = [\text{Re}\{\mathbf{s}\}^T, \text{Im}\{\mathbf{s}\}^T]^T$ and

$$\mathbf{A}_d = 2 [\text{Re}\{\text{diag}(\mathbf{b}_d^*) \hat{\mathbf{G}}_d\} \quad -\text{Im}\{\text{diag}(\mathbf{b}_d^*) \hat{\mathbf{G}}_d\}]. \quad (63)$$

Algorithm 1 Joint DOA and Channel-Gain Estimation

Require: Training observations $\mathbf{z}_p$, pilots $\mathbf{x}_p[t]$, reference vector $\mathbf{b}_p$, RIS geometry, feasible parameter region Ω , maximum iteration number I_c , initial damping factor $\lambda^{(0)}$, and constants $\epsilon_c, \epsilon_0 > 0$.

Ensure: Estimated parameter vector $\hat{\boldsymbol{\eta}}$ and channel $\hat{\mathbf{H}}$.

- 1: Obtain $\mathbf{H}^{(0)}$ from the four-phase field estimate when available; otherwise use (60).
 - 2: Obtain coarse DOAs using (61).
 - 3: Form $\boldsymbol{\eta}^{(0)}$ according to (50).
 - 4: **for** $i = 0, 1, \dots, I_c - 1$ **do**
 - 5: Compute $\mathbf{u}_p^{(i)} = \mathbf{g}_p(\boldsymbol{\eta}^{(i)}) + \mathbf{b}_p$.
 - 6: Compute the residual $\mathbf{e}_p^{(i)} = \mathbf{z}_p - |\mathbf{u}_p^{(i)}|^2$.
 - 7: Construct $\mathbf{J}_p^{(i)}$ using (56).
 - 8: Compute $\Delta \boldsymbol{\eta}^{(i)}$ using (57).
 - 9: Compute the projected trial point using (58).
 - 10: **if** the trial point decreases F_p **then**
 - 11: Accept $\boldsymbol{\eta}^{(i+1)} = \boldsymbol{\eta}_{\text{trial}}^{(i+1)}$.
 - 12: Reduce the damping parameter.
 - 13: **else**
 - 14: Set $\boldsymbol{\eta}^{(i+1)} = \boldsymbol{\eta}^{(i)}$.
 - 15: Increase the damping parameter.
 - 16: **end if**
 - 17: **if** $\|\Delta \boldsymbol{\eta}^{(i)}\|_2 / (\|\boldsymbol{\eta}^{(i)}\|_2 + \epsilon_0) < \epsilon_c$ **then**
 - 18: **break**
 - 19: **end if**
 - 20: **end for**
 - 21: Set $\hat{\boldsymbol{\eta}} = \boldsymbol{\eta}^{(i+1)}$.
 - 22: Reconstruct $\hat{\mathbf{H}} = \mathbf{H}(\hat{\boldsymbol{\eta}})$.
-

A regularized real-domain initialization is

$$\bar{\mathbf{s}}_{\text{lin}} = (\mathbf{A}_d^T \mathbf{A}_d + \gamma_{\text{lin}} \mathbf{I})^{-1} \mathbf{A}_d^T (\mathbf{z}_d - |\mathbf{b}_d|^2), \quad (64)$$

where $\gamma_{\text{lin}} \geq 0$. The complex estimate $[\mathbf{I}; j\mathbf{I}] \bar{\mathbf{s}}_{\text{lin}}$ is projected element-wise onto $\mathcal{S}$ and used either to initialize an iterative detector or to center a reduced ML candidate list.

3) *Reference-Assisted Gerchberg–Saxton Detection*: The conventional Gerchberg–Saxton algorithm alternates between a measurement-magnitude constraint and a signal-domain constraint. The detector considered here is not a direct application of the conventional homogeneous phase-retrieval algorithm. Instead, it is adapted to the affine observation model by including the known reference vector in the forward projection and subtracting it before the signal-domain back-projection.

Given the current symbol estimate $\mathbf{s}^{(i)}$, the predicted affine field is

$$\mathbf{u}_d^{(i)} = \hat{\mathbf{G}}_d \mathbf{s}^{(i)} + \mathbf{b}_d. \quad (65)$$

The measured-magnitude constraint is then enforced as

$$\tilde{\mathbf{u}}_d^{(i)} = \sqrt{[\mathbf{z}_d]_+} \odot \exp(j \angle \mathbf{u}_d^{(i)}), \quad (66)$$

where $[\mathbf{z}]_+ = \max(\mathbf{z}, \mathbf{0})$ is applied element-wise.

The reference vector is then removed, and the resulting field is projected back onto the signal domain. To improve

Algorithm 2 Reference-Assisted Gerchberg–Saxton Symbol Detection

Require: Data observations $\mathbf{z}_d$, matrix $\hat{\mathbf{G}}_d$, reference vector $\mathbf{b}_d$, constellation $\mathcal{S}$, regularization factor γ , maximum iteration number I_d , and constants $\epsilon_d, \epsilon_0 > 0$.

Ensure: Detected symbol vector $\hat{\mathbf{s}}_d$.

- 1: Compute $\mathbf{r}_d = \sqrt{[\mathbf{z}_d]_+}$.
 - 2: Precompute $\mathbf{B}_d$ according to (67).
 - 3: Initialize $\mathbf{s}^{(0)}$ using a linearized estimate, the previous data block, or a random constellation vector.
 - 4: **for** $i = 0, 1, \dots, I_d - 1$ **do**
 - 5: Compute the predicted affine field using (65).
 - 6: Apply the magnitude constraint using (66).
 - 7: Back-project using (68).
 - 8: Apply the constellation projection using (69).
 - 9: **if** $\|\mathbf{s}^{(i+1)} - \mathbf{s}^{(i)}\|_2 / (\|\mathbf{s}^{(i)}\|_2 + \epsilon_0) < \epsilon_d$ **then**
 - 10: **break**
 - 11: **end if**
 - 12: **end for**
 - 13: Output $\hat{\mathbf{s}}_d = \mathcal{Q}_S(\mathbf{s}^{(i+1)})$.
-

numerical stability when $\hat{\mathbf{G}}_d$ is ill-conditioned, define the regularized back-projection matrix

$$\mathbf{B}_d = (\hat{\mathbf{G}}_d^H \hat{\mathbf{G}}_d + \gamma \mathbf{I})^{-1} \hat{\mathbf{G}}_d^H, \quad (67)$$

where $\gamma \geq 0$ is a regularization parameter. When $\gamma = 0$ and $\hat{\mathbf{G}}_d$ has full column rank, $\mathbf{B}_d = \hat{\mathbf{G}}_d^\dagger$.

The continuous signal-domain update is

$$\tilde{\mathbf{s}}^{(i+1)} = \mathbf{B}_d (\tilde{\mathbf{u}}_d^{(i)} - \mathbf{b}_d). \quad (68)$$

The finite-alphabet structure is incorporated through

$$\mathbf{s}^{(i+1)} = \mathcal{Q}_S(\tilde{\mathbf{s}}^{(i+1)}), \quad (69)$$

where $\mathcal{Q}_S(\cdot)$ denotes element-wise nearest-neighbor quantization onto the constellation $\mathcal{S}$.

In the proposed detector, (69) is applied after every back-projection to exploit the finite-alphabet prior. The complete procedure is summarized in Algorithm 2.

The proposed reference-assisted GS detector is attractive for repeated data detection because $\mathbf{B}_d$ can be precomputed whenever the estimated channel remains unchanged over a coherence block. Each subsequent iteration then mainly involves matrix-vector multiplications and element-wise operations.

D. Computational Complexity and Implementation Considerations

Let I_c and I_d denote the numbers of channel-estimation and symbol-detection iterations, respectively. The principal computational costs are summarized in Table I.

For joint DOA and gain estimation, the dominant operations are forming $\mathbf{J}_p^T \mathbf{J}_p$ and solving a $D_\eta \times D_\eta$ system. Since $D_\eta = 4L$, this dimension is independent of N_r and is much smaller than the $2N_r L$ -dimensional unstructured alternative. For GS, forming $\mathbf{B}_d$ costs $\mathcal{O}(Q_d D_s^2 + D_s^3)$ once per coherence block; subsequent iterations require only matrix-vector and element-wise operations. The linearized estimate or the preceding data block can also provide a deterministic warm start.

E. Cramér–Rao Lower Bound for DOA Estimation

The CRLB follows directly from the parametric training model in (51) and its Jacobian in (56). Assume that the effective post-detection noise is white Gaussian, $\mathbf{w}_p \sim \mathcal{N}(\mathbf{0}, \sigma_p^2 \mathbf{I})$, with variance independent of the unknown parameters. Repartition the real parameter vector in (50) as $\boldsymbol{\eta} = [\boldsymbol{\psi}^T, \boldsymbol{\nu}^T]^T$, where $\boldsymbol{\psi} = [\boldsymbol{\theta}^T, \boldsymbol{\phi}^T]^T$ contains the DOAs and $\boldsymbol{\nu} = [\text{Re}\{\boldsymbol{\alpha}\}^T, \text{Im}\{\boldsymbol{\alpha}\}^T]^T$ contains the nuisance path gains. The Fisher information matrix (FIM) is then

$$\mathbf{F}(\boldsymbol{\eta}) = \frac{1}{\sigma_p^2} \mathbf{J}_p^T(\boldsymbol{\eta}) \mathbf{J}_p(\boldsymbol{\eta}), \quad (70)$$

with the column partition $\mathbf{J}_p = [\mathbf{J}_\psi \ \mathbf{J}_\nu]$. Eliminating the unknown gains gives the equivalent FIM for the DOAs,

$$\begin{aligned} \mathbf{F}_{\text{DOA}} &= \mathbf{F}_{\psi\psi} - \mathbf{F}_{\psi\nu} \mathbf{F}_{\nu\nu}^\dagger \mathbf{F}_{\nu\psi} \\ &= \frac{1}{\sigma_p^2} \mathbf{J}_\psi^T \mathbf{P}_\nu^\perp \mathbf{J}_\psi, \\ \mathbf{P}_\nu^\perp &= \mathbf{I} - \mathbf{J}_\nu \mathbf{J}_\nu^\dagger, \end{aligned} \quad (71)$$

where $(\cdot)^\dagger$ denotes the Moore–Penrose inverse. Thus, only the component of an angle-induced intensity perturbation orthogonal to the gain perturbation subspace carries DOA information. If $\mathbf{F}_{\text{DOA}}$ is nonsingular, any unbiased DOA estimator satisfies

$$\text{Cov}(\hat{\boldsymbol{\psi}}) \succeq \mathbf{F}_{\text{DOA}}^{-1}. \quad (72)$$

The diagonal entries give the individual bounds for θ_l and ϕ_l . Thus, weak or rank-deficient directions of $\mathbf{P}_\nu^\perp \mathbf{J}_\psi$ respectively produce large or unbounded DOA limits, linking reference design directly to attainable angular accuracy.

V. SIMULATION RESULTS

Numerical results evaluate channel estimation, symbol detection under perfect and estimated channel state information (CSI), and the effects of reference and RIS measurement diversity.

Unless otherwise stated, the simulations follow the two-phase model in Section II with a two-user 8×8 holographic RIS operating at 3.5 GHz and half-wavelength element spacing. The complex path gains are $\alpha_1 = e^{j0.35}$ and $\alpha_2 = 0.85e^{-j0.55}$, normalized QPSK is used, and the reference amplitude is 1.5. Training and data observations follow (51) and (12) with independent real Gaussian post-detection noise of variance σ_w^2 per intensity sample. For a noiseless intensity vector $\boldsymbol{\mu} = |\mathbf{G}\boldsymbol{\vartheta} + \mathbf{b}|^2 \in \mathbb{R}^Q$, the SNR is defined as

$$\text{SNR} = 10 \log_{10} \left(\frac{\mathbb{E}\{\|\boldsymbol{\mu}\|_2^2\}}{Q\sigma_w^2} \right). \quad (73)$$

Scenario-specific DOAs, pilot lengths, modulation orders, and reference or RIS configurations are stated with the corresponding experiments.

A. Channel Estimation Performance

Unless stated otherwise, this subsection uses $T_p = 32$ QPSK pilot observations and user DOAs $(15^\circ, 8^\circ)$ and $(35^\circ, -10^\circ)$, and the path-gain magnitudes are 1 and 0.85, respectively. Each point is averaged over 30 independent noise realizations

TABLE I
COMPUTATIONAL COMPLEXITY OF THE MAIN RECOVERY ALGORITHMS

Method	Unknown dimension	Approximate complexity	Primary role
Joint damped GN	D_η	$\mathcal{O}(I_c (Q_p D_\eta^2 + D_\eta^3))$	DOA and channel-gain estimation
Exhaustive ML	D_s	$\mathcal{O}(S ^{D_s} Q_d D_s)$	Performance benchmark
Linearized detector	D_s	$\mathcal{O}(Q_d D_s^2 + D_s^3)$	Initialization and candidate screening
Reference-assisted GS	D_s	$\mathcal{O}(I_d Q_d D_s)$	Low-complexity symbol detection

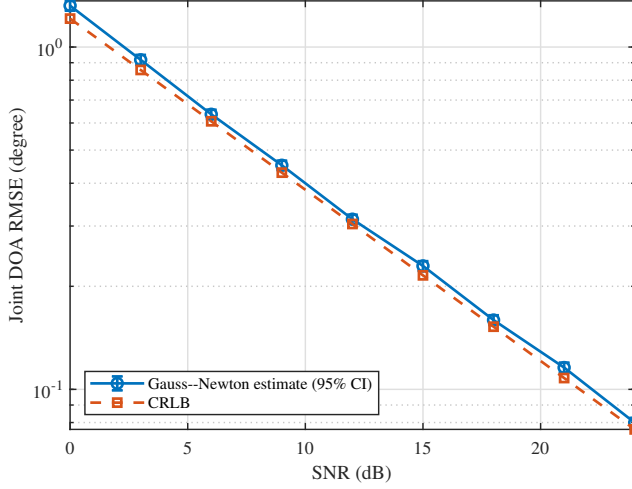


Fig. 3. Joint DOA RMSE and the corresponding CRLB versus SNR.

unless noted otherwise. The reference phase pattern is selected from a 32-entry codebook by maximizing the minimum singular value of the training Jacobian.

Based on 1000 noise realizations, Fig. 3 shows that the joint DOA RMSE decreases from 1.32° at 0 dB to 0.08° at 24 dB and closely follows the CRLB. Because the constrained nonlinear estimator can be biased at finite SNR, the ordinary unbiased-estimator CRLB is used as a local accuracy benchmark rather than a strict finite-sample bound.

Figure 4 examines the counting hierarchy in (32) for the per-element $L = 2$ unstructured training problem. All operating points use prefixes of the same QPSK pilot sequence and 10,000 paired trials. At $T_p = 2L$, regular local recovery becomes possible but the noiseless multistart success probability is only about 0.54; at $T_p = 3L$, it exceeds 0.99 for every nonzero reference design. The no-reference model retains its global phase ambiguity and an approximately 3 dB NMSE floor. Beyond the identifiability threshold, additional measurements improve noise robustness, reaching about -19.3 dB at $T_p = 48$. Success for an individual instance below $3L$ does not replace the uniform global condition.

In Fig. 5, the post-detection noise variance is fixed at its nominal $\rho = 1.5$ value. The reference lowers the NMSE from 3.12 dB at $\rho = 0$ to -17.39 dB at $\rho = 1.5$ by strengthening the interference term. This idealized trend excludes saturation and quantization and therefore does not imply unbounded hardware gains.

Figure 6, averaged over 300 paired trials, compares equal-power reference patterns. Without a reference, the singular training Jacobian ($\sigma_{\min} = 3.69 \times 10^{-16}$) produces a persistent 3 dB error floor. The reference-assisted curves instead decrease

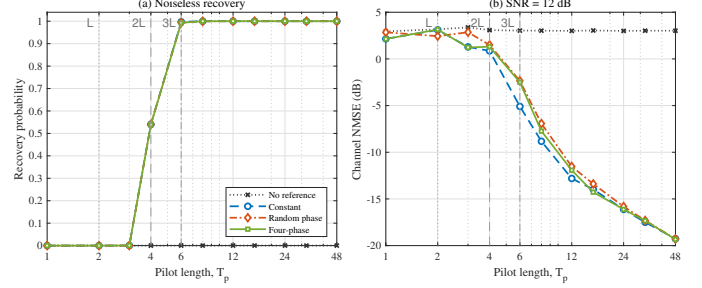


Fig. 4. Effect of pilot count and reference design for the two-user unstructured training problem: (a) empirical noiseless multistart recovery probability and (b) channel NMSE at 12 dB. The vertical lines mark $T_p = L, 2L$, and $3L$.

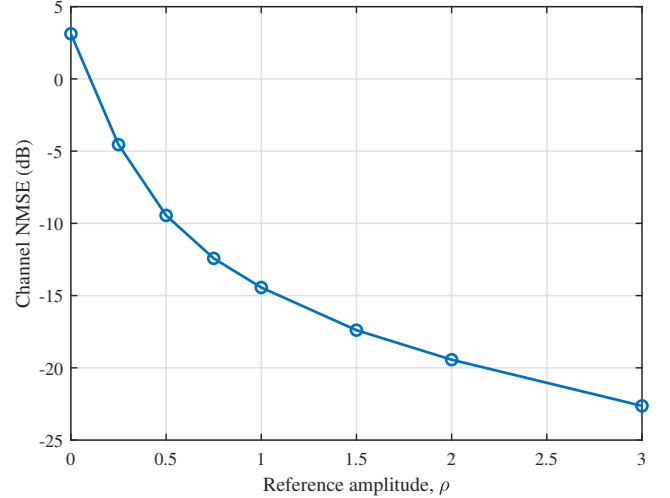


Fig. 5. Channel-estimation NMSE versus reference amplitude for fixed post-detection noise variance.

with SNR; the uniform pattern gives the largest minimum singular value (9.95) and is also selected by the codebook search. Thus, the reference removes the phase ambiguity, while its phase design further improves conditioning.

B. Symbol Detection Performance with Perfect CSI

We first isolate the data detector from channel-estimation errors by constructing $\mathbf{G}_d$ from the true channel. Unless stated otherwise, two QPSK users at the DOAs used in the preceding subsection, four reference states, and a reference amplitude of 1.5 are considered.

Figure 7 compares the detectors under the codebook-selected reference. Each SNR point is averaged over 10^5 independent trials, corresponding to 2×10^5 user-symbol decisions. Projected GN closely follows the ML benchmark, whereas GS incurs a moderate alternating-projection loss but

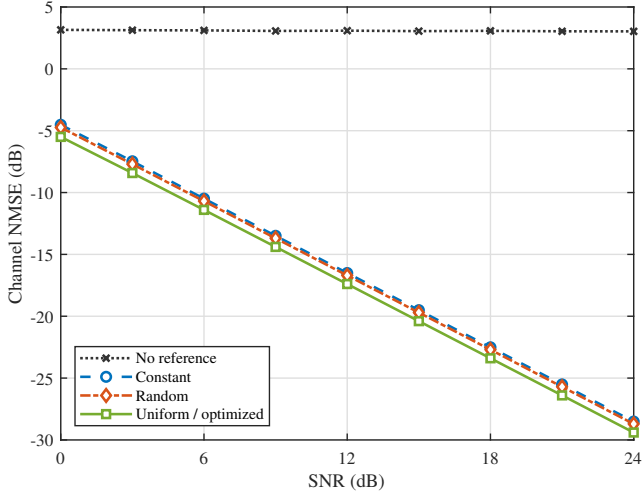


Fig. 6. Channel-estimation NMSE versus SNR for different training-reference designs.

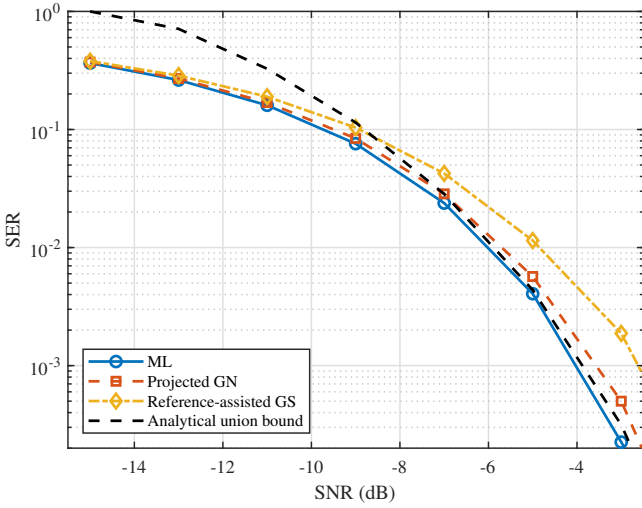


Fig. 7. SER of different symbol detectors under perfect CSI, averaged over 10^5 trials per SNR, and the analytical distance-spectrum union bound.

retains the same waterfall trend; at -5 dB their SERs are 4.07×10^{-3} , 5.69×10^{-3} , and 1.15×10^{-2} , respectively. The analytical SER union bound, obtained by summing the pairwise error probabilities in Corollary 2 with symbol-error multiplicities, upper-bounds the ML result and becomes tight in the waterfall region; at -5 dB, the bound is 4.38×10^{-3} .

Figure 8 compares equal-power references under the energy-domain SNR in (73). Without a reference, $d_{\min}^{(E)} = 0$ and the QPSK SER remains near 0.75. All reference-assisted designs remove this ambiguity. The uniform pattern has the largest normalized minimum distance (9.79) and is selected by the codebook search. Each point contains 40000 user-symbol decisions; zero-error points are plotted at the half-event level 1.25×10^{-5} .

To test whether this distance predicts more than the five representative designs, Fig. 9 uses a 32-entry equal-power reference codebook with different phase and spatial-support patterns. At -6 dB, the correlation between the normalized minimum distance and the logarithm of the ML SER is

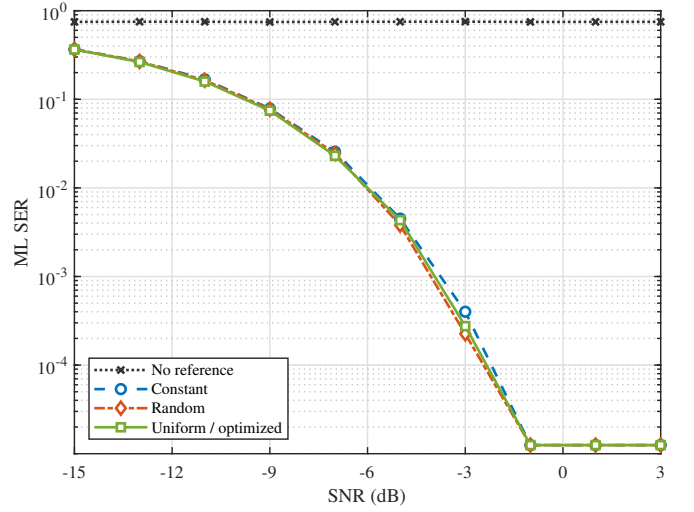


Fig. 8. ML SER versus SNR for different data-phase reference designs under perfect CSI. Zero-error points are displayed at the half-event level.

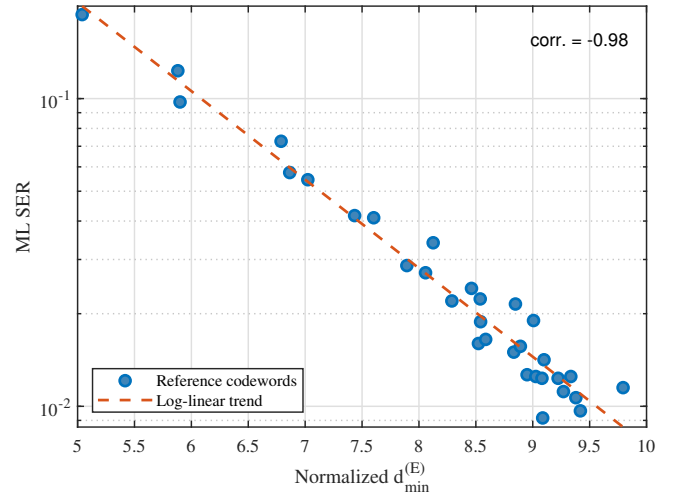


Fig. 9. ML SER versus normalized energy-domain minimum distance over an equal-power reference codebook.

-0.98 . Thus, increasing the worst-case intensity separation systematically lowers the detection error, directly supporting the max-min criterion in (49).

Finally, Fig. 10 uses the same optimized reference to compare QPSK, 16-QAM, 64-QAM, and 256-QAM. As the modulation order increases, the denser constellation reduces the separation between intensity codewords and shifts the SER curve toward higher SNR. Over the sampled SNR grid, the SER first falls below 10^{-2} at -3 , 3 , 9 , and 15 dB for QPSK, 16-QAM, 64-QAM, and 256-QAM, respectively. Thus, the reference removes the structural phase ambiguity for all four constellations, whereas the remaining SNR requirement is governed by finite-alphabet energy separation.

C. End-to-End Symbol Detection with Estimated CSI

Figure 11 constructs $\mathbf{G}_d$ from the estimated channel and uses equal training and data SNRs. At -6 dB, estimated-CSI SERs are about 0.2, compared with roughly 10^{-2} under

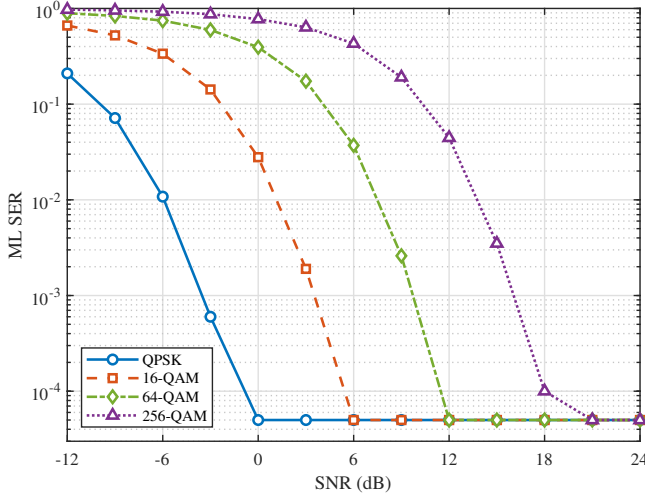


Fig. 10. ML SER for QPSK, 16-QAM, 64-QAM, and 256-QAM under perfect CSI. Zero-error points are displayed at the half-event level.

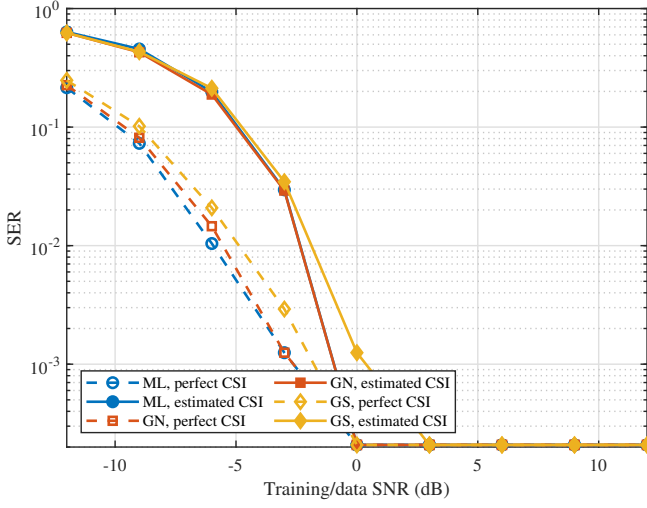


Fig. 11. SER under perfect and estimated CSI for different symbol detectors.

perfect CSI, and the gap narrows as the channel NMSE improves. Channel mismatch, rather than the choice among the three detectors, therefore dominates at low SNR.

Figure 12 separates the two stages. With data SNR fixed at -6 dB, increasing the training SNR lowers the reference-assisted SER toward the data-noise floor, whereas the no-reference case remains near 0.75 . With training SNR fixed at 12 dB, the same error floor persists without a data reference, while all phase-diverse designs show the expected waterfall. Thus, the training reference enables CSI recovery and the data reference ensures finite-alphabet separability.

D. Effect of Reference and RIS Measurement Diversity

We finally examine how spatial and reference-state diversity change the finite-alphabet decision margin. To compare configurations with different numbers of observations, $d_{\min}^{(E)}$ is normalized by the root-mean-square noiseless intensity. The two-user QPSK setting and four-phase reference are used unless otherwise stated.

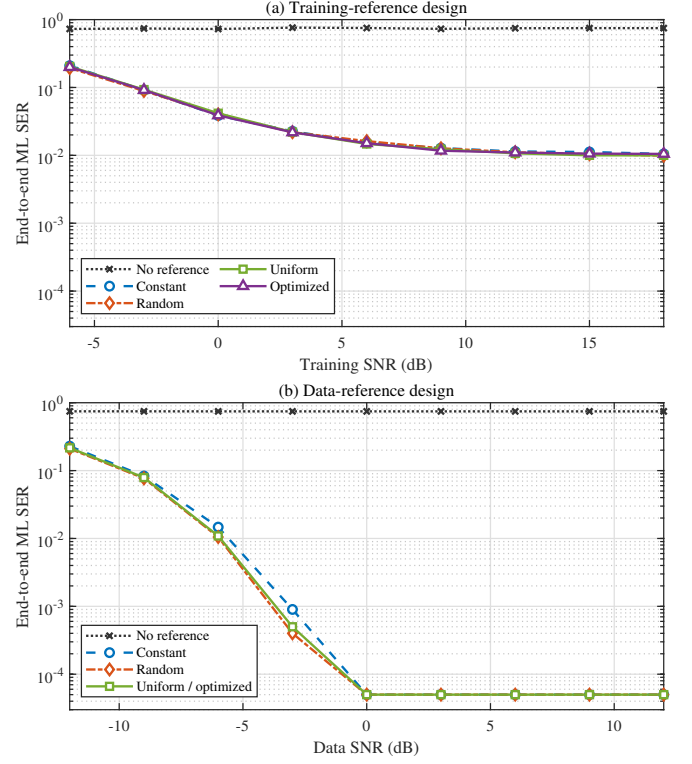


Fig. 12. End-to-end ML SER versus (a) training SNR for different training-reference designs at a data SNR of -6 dB and (b) data SNR for different data-reference designs at a training SNR of 12 dB. Zero-error points are displayed at the half-event level.

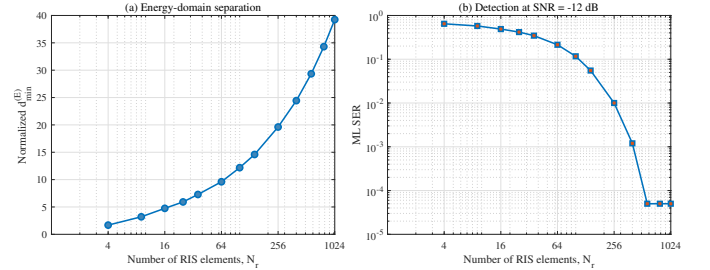


Fig. 13. Effect of the number of RIS element-level observations on (a) the normalized energy-domain minimum distance and (b) ML detection at -12 dB. Zero-error points are displayed at the half-event level.

At -12 dB, Fig. 13 shows that increasing N_r from 4 to 1024 raises the normalized minimum distance from 1.68 to 39.24 and drives the ML SER toward zero. This close correspondence confirms that the element outputs provide useful spatial diversity.

Reference-state diversity gives the analogous trend in Fig. 14. Uniformly spacing R phases over $[0, 2\pi)$ increases the normalized distance from 2.75 to 19.21 and lowers the -12 dB SER from 0.501 to 0.0111 as R grows from 1 to 16 , at the cost of linearly increasing measurement and switching overhead.

Figure 15 connects spatial conditioning to end-to-end detection. Increasing the separation from 0.25° to 8° raises the normalized minimum singular value from 0.0214 to 0.628 and lowers the estimated-CSI SER from 0.259 to 2.57×10^{-2} .

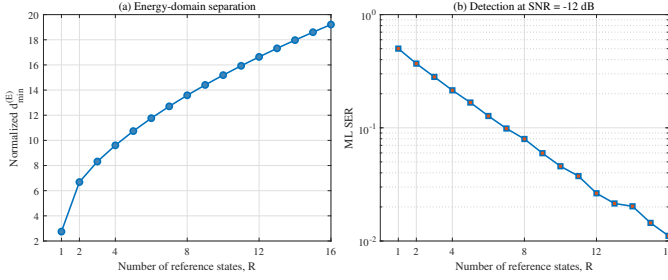


Fig. 14. Effect of the number of phase-shifted reference states on (a) the normalized energy-domain minimum distance and (b) ML detection.

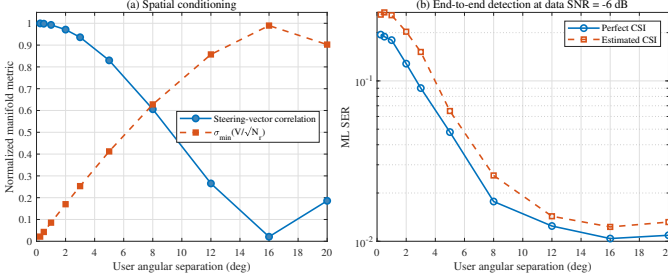


Fig. 15. Effect of user angular separation on (a) array-manifold conditioning and (b) end-to-end ML detection.

The small rebound beyond 16° reflects finite-aperture side-lobes. Here the training and data SNRs are 12 and -6 dB, respectively, with $T_p = 16$.

Finally, Fig. 16 compares equal-overhead designs with 64 samples. Selecting state-dependent element masks from a 4096-entry codebook increases the normalized minimum distance from 4.75 to 4.84. Both ML and GS exhibit the same modest but consistent improvement, showing that RIS sampling can complement reference-phase diversity.

These results support the two-part design principle in Section III: the RIS measurement matrix must first preserve relevant field differences, after which reference-phase diversity projects those differences into separated intensity observations.

E. Effect of the Reference-to-Signal Energy Ratio

We finally examine the energy balance identified in (24). A single received field snapshot over the same 8×8 RIS is measured using the four uniformly spaced reference phases, while the total average incident energy per element is fixed as $P_s + P_b = 1$. Hence, sweeping the reference-to-signal energy ratio $\text{RSR} = 10 \log_{10}(P_b/P_s)$ changes only the allocation between the unknown and reference fields, rather than the detector power budget. Each point in Fig. 17 uses 5000 noise realizations and direct four-phase field reconstruction.

As predicted by (25), Fig. 17(a) shows that both the normalized interference depth κ and the relative phase information κ^2 peak at the balanced point $\text{RSR} = 0$ dB and decay when either field dominates. The phase RMSE in Fig. 17(b) consequently has a bowl-shaped dependence on RSR and closely follows the phase CRLB in the well-conditioned region. At 20 dB, its minimum is 4.98° , and it remains below 10° for sampled RSRs from -12 to 12 dB. The corresponding intervals are

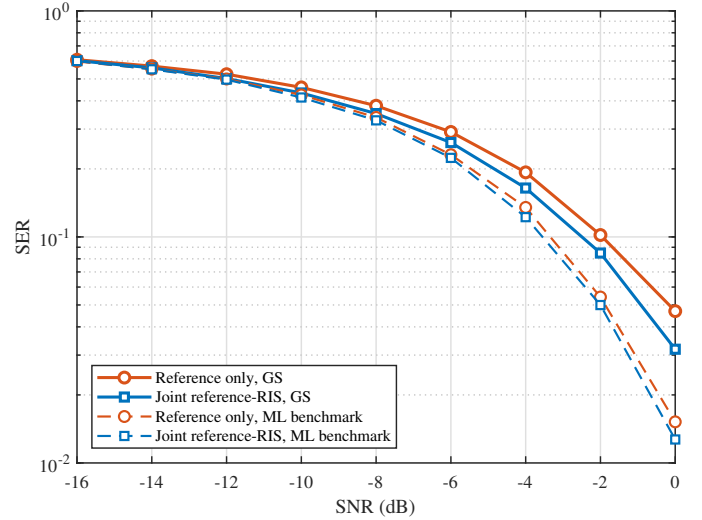


Fig. 16. SER of the proposed reference-assisted GS detector and the ML benchmark under equal-overhead reference-only and joint reference-RIS measurement diversity.

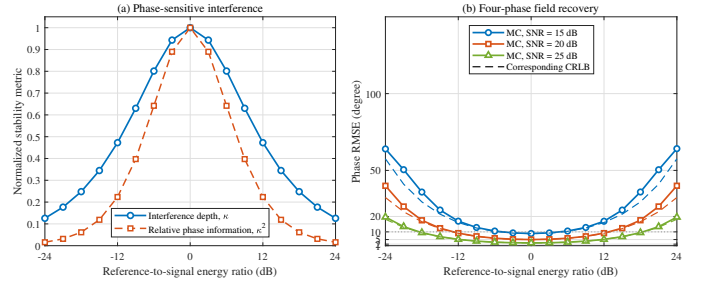


Fig. 17. Effect of the reference-to-signal energy ratio under a fixed total incident-energy budget: (a) normalized interference depth and relative phase information and (b) phase RMSE of four-phase field recovery. Solid curves with markers are Monte Carlo results, dashed curves are the corresponding CRLBs, and the horizontal dotted line denotes a 10° RMSE criterion.

$[-3, 3]$ dB at 15 dB SNR and $[-18, 18]$ dB at 25 dB SNR. Thus, the useful energy-ratio range is SNR dependent rather than universal; nevertheless, severe imbalance consistently weakens phase recovery.

VI. CONCLUSION

This paper developed a reference-assisted holographic RIS receiver that recovers complex channel and data information from intensity-only measurements. The exact secant identity and energy-domain minimum distance connected continuous channel identifiability, finite-alphabet uniqueness, and noise robustness, while the four-phase construction provided an explicit stability guarantee. A damped Gauss-Newton estimator and a reference-assisted GS detector implemented the training and data stages, respectively, with the CRLB and ML detector serving as benchmarks. Simulations confirmed that reference phase design, pilot diversity, array size, and angular separation jointly determine estimation and detection performance. Future work will incorporate hardware-dependent noise, quantization and dynamic-range constraints, and frequency-selective channels.

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